

BOAMP Digital Procurement Study

Successor Linkage, Survival, and Trend Methodology

BOAMP Data Science Internship Project

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Technical Summary

This report states the BOAMP successor-linkage, survival, and trend methodology and the results it produces. The study does not claim to certify legal contract renewals. Its event is an *observable successor procurement*: a later BOAMP procurement episode from the same buyer that is sufficiently similar to an earlier awarded digital procurement episode.

The Grand Ouest survival cohort contains 3,800 awarded digital procurement episodes. The main linkage rule accepts 544 successor links, giving an observed event rate of 0.143 and a median observed successor time of 31.82 months among linked events. The candidate pool is broad: 763,417 candidate pairs from 3,520 anchors with at least one candidate.

Linkage accuracy is read against the Grand Ouest regional reference: a stratified review of 120 anchors of which 88 are usable, carried out against real BOAMP notices before these methods existed. On its locked split of 72 anchors (18 with a reviewed successor), `M_B_text_ranking @ 0.70` has precision 0.875 (95% CI 0.529–0.978), recall 0.389 (95% CI 0.203–0.614), false-positive rate 0.000, and 8 accepted links. Recall is capped at 0.913 by candidate generation, which reaches 21 of the 23 reviewed successors. These are reference-sample estimates on fewer than a hundred anchors, not independently validated accuracy: the labels were generated by a single LLM research pass and spot-checked on a subset by the project owner, not verified anchor-by-anchor and not reviewed by an independent specialist panel. `M_C_weighted_gated` recovers more positives (recall 0.667) but admits more false positives (FPR 0.185). This trade-off is important because a false positive in a survival dataset creates both a false event and a false event time, while an abstained link is handled conservatively as right-censoring.

1 Context

Gigalis is the digital central purchasing body (*centrale d’achat numérique*) for public institutions in the Pays de la Loire region and beyond. It designs pooled framework agreements in cloud, cybersecurity, digital infrastructure, and artificial intelligence, and its strategic relevance depends on anticipating members’ future needs. The internship’s central question is: using historical public digital procurement data, how can the probability that a contract or technological segment generates an identifiable purchasing need within the next 12–24 months be estimated, and how can that estimate inform Gigalis’s central purchasing strategy?

This decomposes into three sub-problems: a *lifetime problem* (survival analysis: how long between a digital procurement episode and its successor?), a *trend problem* (change-point detection: which technological segments are growing, plateauing, or declining?), and a *text-signal problem* (NLP: extracting a contract’s technological theme automatically). This project answers the first two in depth; the NLP sub-problem is addressed by a supervised technology classifier trained on 500 annotated notices, layered over the reproducible CPV segmentation rather than replacing it (§8).

BOAMP does not encode legal contract renewal as an explicit field, so the study cannot certify legal renewals. It instead constructs and evaluates a proxy: an *observable successor procurement*, a later BOAMP episode from the same buyer that is sufficiently similar to, and temporally plausible relative to, an earlier awarded digital procurement episode. Every downstream result – survival curves, hazard ratios, trend segments – is conditioned on this proxy, not on certified legal ground truth. The strongest defensible framing is: because legal renewal status is not directly observed, the study measures observable successor procurements under a precision-first linkage rule, then assesses how sensitive its descriptive conclusions are to that rule.

2 What The Pipeline Measures

Let i index an earlier awarded procurement episode (the *anchor*) and j index a later candidate episode. The object of interest is not a legal renewal certificate, because BOAMP notices do not provide that ground truth consistently. Instead, the pipeline estimates whether the data show a later procurement episode that is credible enough to be treated as a successor:

$$Y_i = \begin{cases} 1, & \text{if one accepted observable successor is found before the cutoff,} \\ 0, & \text{otherwise.} \end{cases}$$

The event time is

$$\tau_i = v_{\hat{j}} - u_i,$$

where u_i is the award-date origin of episode i , $v_{\hat{j}}$ is the first-publication date of the accepted successor, and $\hat{j}$ is the selected candidate. If no accepted successor is found before 2025-12-31, the episode is right-censored.

2.1 Notation

Three layers are chained – candidate generation, linkage decision, survival – and each answers a different conditional question. One symbol table serves the whole report so that no quantity below is undefined at the point it is used.

Layer	Symbol	Meaning
Data	u_i	award-date origin of anchor i (calendar date)
Data	v_j	first-publication date of candidate j (calendar date)
Data	J_i	set of candidates for anchor i surviving blocking
Data	$T_{ij}, B_{ij}, K_{ij}, G_{ij}$	pairwise text, buyer, CPV-continuity and timing evidence
Reference	$R_i \in J_i \cup \{\emptyset\}$	reviewed successor identity, $\emptyset$ if the review found none
Reference	$P_i = \mathbf{1}\{R_i \neq \emptyset\}$	the reference identifies a successor for anchor i
Reference	$E_i = \mathbf{1}\{R_i \in J_i\}$	the reviewed successor survived candidate generation
Decision	$\hat{R}_i \in J_i \cup \{\emptyset\}$	successor accepted by the linkage rule, $\emptyset$ if it abstains
Decision	$A_i = \mathbf{1}\{\hat{R}_i \neq \emptyset\}$	the method accepted some successor
Decision	$C_i = \mathbf{1}\{\hat{R}_i = R_i \neq \emptyset\}$	the accepted successor is exactly the reviewed one
Decision	$\mathcal{L}_m$	one linkage definition (method and threshold), m indexing the arms
Survival	T_i	months from award to an observable successor (latent)
Survival	L_i	follow-up months from award to the 2025-12-31 cutoff
Survival	$Y_i = \mathbf{1}\{T_i \leq L_i\}$	an accepted observable successor was seen before the cutoff
Survival	$S(t), F(t), \lambda(t)$	survivor, cumulative-event and hazard functions of T
Survival	a, h	current episode age and forward horizon, in months

Table 1: Symbols used throughout, grouped by the layer they belong to. Reading across the three middle blocks gives the logic of the study: blocking decides whether $E_i = 1$, the linkage rule decides A_i and hence C_i , and the survival layer turns A_i into the event indicator Y_i .

Two conventions are worth stating because they prevent the two collisions a reader would otherwise hit. Calendar dates are lower-case (u_i, v_j) so that the upper-case A_i and C_i can carry their usual meaning as decision indicators. The hazard is written $\lambda(t)$ rather than the equally common $h(t)$, because h is already the forward horizon in the operational quantity $P(T \leq a + h \mid T > a)$, which is the report’s headline number. Pairwise evidence always carries two indices (T_{ij} is a text similarity); a single index means an episode-level quantity (T_i is a survival time).

2.2 One Chain Of Questions

The methods below are not a collection of separate exercises. Each stage hands the next one its input, and each is best remembered by the question it answers.

Step	Quantity	Question it answers
1	$P(E = 1 \mid P = 1)$	If a reviewed successor exists, did blocking keep it reachable?
2	$P(C = 1 \mid A = 1)$	If the method accepts a link, is the accepted candidate the reviewed one?
2	$P(C = 1 \mid P = 1)$	If a reviewed successor exists, is it recovered exactly?
3	$Y_i = \mathbf{1}\{A_i = 1\}$	Does this episode contribute an event or censored exposure?
4	$S(t) = P(T > t)$	How long do episodes go without an observable successor?
5	$P(T \leq a + h \mid T > a)$	Given none by age a , how likely is one in the next h months?
6	$\lambda(t \mid X) = \lambda_0(t)e^{\beta^\top X}$	Which episode characteristics go with a sooner successor?
7	$P_m(T \leq t), \text{HR}_{k,m}$	Which of those conclusions survive a change of linkage definition?
8	PELT, HMM, OLS	When did the market’s level shift, and where is it now?

Table 2: The study as one chain. Steps 1–2 measure the instrument, step 3 builds the data, steps 4–6 estimate, step 7 stress-tests, step 8 describes the surrounding market.

3 End-to-End Pipeline

The implemented pipeline is:

BOAMP notices $\rightarrow$ standardised notices $\rightarrow$ procurement episodes
 $\rightarrow$ candidate pairs $\rightarrow$ linkage scoring $\rightarrow$ one successor or abstention $\rightarrow$ survival dataset.

Standardisation and feature engineering. The preprocessing stage standardises dates, buyer identifiers, buyer names, CPV codes, text fields, procedure information, framework flags, and duration fields where they are explicitly available. It does not impute missing duration or invent expected expiry dates. Buyer standardisation is deliberately conservative: municipal variants such as commune/ville/mairie can be harmonised, but intercommunal legal forms are preserved as distinct entities, and conflicting validated SIRENs block a buyer match. The legal-form audit reports 0 accepted links with conflicting validated SIREN evidence and 0 accepted municipal/intercommunal mixes.

Episode reconstruction. BOAMP is notice-level data, but the analysis needs procurement-level events. Therefore notices belonging to the same procurement are reconstructed into an episode. This avoids treating a consultation notice, correction notice, award notice, or lot-level administrative notice as separate renewals.

Candidate generation. Candidate generation is intentionally broad and is not the final model. For an anchor episode with award date u_i and candidate publication date v_j , the candidate enters the exposed set J_i only if

$$u_i + 90 \leq v_j \leq u_i + 2920.$$

The lower bound removes very near follow-up notices and parallel administrative activity. The upper bound is approximately eight years; it keeps the candidate pool inclusive enough for long public-procurement cycles. Precision is then controlled by the selection stage rather than by a narrow time window. The blocking rule generates 763,417 candidate pairs, with a median of 89 candidates per anchor.

The question this stage must answer is *not* how often it is right. It is whether a successor that genuinely exists can still be found downstream, which is the conditional probability

$$P(E_i = 1 \mid P_i = 1) \quad \text{estimated by} \quad \hat{P}(E = 1 \mid P = 1) = \frac{21}{23} = 0.913.$$

In words: given that the reference identifies a successor for an anchor, blocking retained that successor in 21 of the 23 reviewed cases. This is candidate-generation reachability measured on this reference sample – in record-linkage terms *pairs completeness* – and it is not an estimate of population recall. It is a hard ceiling: a successor discarded here can never be recovered by any scorer, which is why the two stages are given opposite objectives. Candidate generation maximises $P(E = 1 \mid P = 1)$; the linkage rule then maximises $P(C = 1 \mid A = 1)$ (§10). High recall first, high precision later.

That objective is why exact CPV continuity is *not* imposed as a hard blocking rule, even though it would raise precision: CPV coding is incomplete, often generic, and assigned by the contracting authority rather than validated, so a same-division requirement would remove reviewed successors from J_i and lower $P(E = 1 \mid P = 1)$ permanently. Christen and Goiser (2007) identify pairs completeness as a confounder that must be published alongside any linkage-quality figure rather than folded into it, which is why it is reported here as its own conditional probability rather than absorbed into recall.

Relaxing hard same-CPV blocking is checked against the reference rather than asserted. Among the 23 reviewed successor pairs – labelled with no knowledge of any linkage method – 9 (39.1%) connect episodes in *different* CPV divisions. A hard same-division block would therefore discard those 9 reviewed successors outright and cut the attainable recall ceiling to 0.609. The discarded cases are substantive continuations rather than noise: equipment purchases followed by their maintenance contracts (CPV-32 or CPV-35 to CPV-50) and software purchases followed by the corresponding service contracts (CPV-48 to CPV-72). This is the ordinary goods/services split built into the CPV hierarchy itself, compounded by assignment error that Siciliani et al. (2023) report is frequent even for human experts given the size of the vocabulary.

4 Grand Ouest Regional Reference

Linkage evaluation uses a Grand Ouest regional reference sample drawn from the study population itself: 120 awarded digital procurement anchors stratified by CPV theme, buyer-identifier quality, and duration availability across Bretagne, Pays de la Loire, and Normandie. Each anchor was reviewed on 2026-08-11 against the notices and official BOAMP URLs of its candidates. The review is a single LLM research pass over those notices, their official URLs, and wider public sources, spot-checked on a subset by the project owner; it proposes one successor or an abstention per anchor. No linkage method existed or was consulted when the labels were produced, so the labels are independent of every method scored against them.

Reference anchors are resolved onto the procurement-episode table through their BOAMP notice identifiers. 8 anchors did not resolve to exactly one episode and were dropped rather than guessed, and anchors the review declined to decide are excluded rather than counted as negatives, leaving 88 evaluable anchors.

The sample is split into a pilot part and a locked part. The acceptance threshold was fixed before the locked part was read, which is what allows locked-split figures to be reported as held out. The pilot split carries 16 evaluable anchors (5 with a reviewed successor) and 5,221 pair rows; the locked split carries 72 evaluable anchors (18 with a reviewed successor) and 20,917 pair rows.

Two anchor counts appear for the locked split and the difference is by construction, not a discrepancy: 72 anchors are evaluable at anchor level, of which 69 also have at least one exposed candidate pair and so appear in the pair-level table. The remaining 3 generated no candidate at

all, which is a blocking-stage loss counted against recall rather than a scoring failure. Anchor-level metrics use 72; pair-level ROC and precision-recall curves use 69.

Candidate generation, not scoring, sets the ceiling on recall against this reference: $P(C = 1 \mid P = 1) \leq P(E = 1 \mid P = 1) = 0.913$, as quantified above. No method evaluated here can exceed that, and the gap is a blocking-stage limitation rather than a scoring failure.

Both unreachable cases were attributed to the specific blocking condition that rejected them, evaluated in the order the generator applies them, and neither is an implementation defect. In the first, the reviewed anchor never reached the blocking step at all: its episode carries an award notice, but that notice has no structured Grand Ouest address, so it is absent from the regional notice table, no award date could be resolved, and the episode was dropped from the cohort – one of the 26 episodes the cohort funnel already counts under that heading. In the second, the buyer changed legal form between the two procurements, from a communal social action centre (CCAS) to its intercommunal successor (CIAS); the anchor carries no SIREN, so no identifier could bridge the transition, and the normalised names differ. Buyer blocking rejected the pair by design, because the pipeline deliberately preserves intercommunal legal forms as distinct entities rather than merging them on name similarity.

That second case is the known weak point of buyer identification in French procurement notices rather than a defect specific to this pipeline: [Potin et al. \(2023\)](#) report missing agent identification as the most serious quality problem in the French data, with buyer SIRETs populated on a minority of lots, and note that the same authority appears under alternative and former names. Recovering this particular link would require an external legal-succession register, not a change to the blocking rule. Both losses are accepted and documented rather than repaired; neither is repaired by loosening the rule, and the audit reports 0 unexplained cases, which is the check that distinguishes a blocking trade-off from a bug. The attribution is regenerated by `scripts/audit_candidate_generation.py`.

Five limitations bind everything computed from the regional reference. The labels were spot-checked on a subset rather than verified anchor-by-anchor; anchors outside that subset carry the model’s judgement as recorded, so this is a reference sample and not ground truth, and it is not an independent human specialist panel. The sources behind each individual label were not recorded, so a given anchor’s evidence trail cannot be fully reconstructed or independently re-executed. Negatives are corpus-relative: roughly 25 candidates per anchor were considered rather than the full pool, so a false-positive rate computed on them is conservative by construction and is a diagnostic on this sample rather than a population-wide rate. Judging whether later notice text continues an earlier need draws on the same text, CPV, and date evidence the text-ranking method scores, so the labels are method-independent without being fully evidence-independent. And the sample is small enough that every point estimate needs its interval read beside it.

5 Linkage Algorithms

All methods operate on the same exposed candidate set. This is crucial: the primary method is not comparing text over the whole BOAMP universe. Buyer and time plausibility are imposed before text ranking.

M_A : deterministic evidence. Define B_{ij} as buyer-identity support, K_{ij} as CPV continuity, and T_{ij} as text similarity. The deterministic rule accepts a link only when strong buyer evidence is present, CPV continuity is positive, and a minimum text signal is present:

$$B_{ij} = 1, \quad K_{ij} > 0, \quad T_{ij} \geq t_A.$$

It is interpretable, but it loses recall when CPV or buyer identifiers are missing or noisy.

M_B : text ranking. For each episode, the text fields are converted to TF-IDF vectors x_i and x_j . Text similarity is cosine similarity:

$$T_{ij} = \cos(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\| \|x_j\|}.$$

Within the exposed candidate set J_i , the method selects

$$\hat{j}_i = \arg \max_{j \in J_i} T_{ij},$$

and accepts that candidate as $\hat{R}_i$ only if

$$T_{i\hat{j}_i} \geq 0.70, \quad \text{otherwise} \quad \hat{R}_i = \emptyset.$$

The abstention branch is not a formality: it is what makes A_i a genuine decision and what turns a non-accepted anchor into a censored observation rather than a negative one. This is the primary method because it is simple, reproducible, auditable, and best matches the precision-first objective.

M_C : weighted gated score. This method combines evidence components into a score:

$$S_{ij} = 0.50B_{ij} + 0.25T_{ij} + 0.20K_{ij} + 0.05G_{ij},$$

where G_{ij} is a timing-plausibility score. Missing components are handled by the implemented score-normalisation logic rather than imputation. The method then applies gates and ranks by S_{ij} . It is useful as a higher-recall contrast, but the validation results show a materially higher false-positive rate.

M_D : Fellegi–Sunter probabilistic linkage. Let γ_{ij} be the discretised comparison vector for buyer, text, CPV, and timing evidence. Fellegi–Sunter estimates the likelihood of observing γ_{ij} under a match class M and a non-match class U , then uses the log-likelihood ratio

$$w_{ij} = \log \frac{P(\gamma_{ij} | M)}{P(\gamma_{ij} | U)}.$$

The fitted model provides `fs_match_weight` and `fs_match_probability`. On this cohort it does not outperform the simple text-ranking rule, likely because reviewed successors are rare and same-buyer procurement activity contains many non-renewal lookalikes.

Declared duration. Declared contract duration is not imposed as a mandatory successor-linkage condition. Reliable duration is unavailable for most episodes, so a duration-conditioned rule could differentiate itself only on a minority of the cohort; and where the evidence does exist, the observed relationship between declared duration and successor timing is too weak to support a hard expected-expiry rule – many declared successors are published well before the declared end date. Duration is therefore retained as a descriptive diagnostic about data reliability, not as an event-definition criterion, and the descriptive comparison between declared duration and observed successor delay remains part of the evidence on that basis.

Sensitivity framework. Event-definition sensitivity is carried by four threshold and method arms ($M_B@0.80$, $M_B@0.70$, $M_B@0.60$, $M_C@0.70$), which vary the decision rule along the dimension that actually moves the event set, together with the borderline-band check described in §6.

6 Survival Modeling

Event and censoring. The accepted-link decision from M_B defines the survival event. Let T_i be the months from award to an observable successor and L_i the months from award to the 2025-12-31 cutoff. The pipeline observes

$$\tilde{T}_i = \min(T_i, L_i), \quad Y_i = \mathbf{1}\{T_i \leq L_i\} = \mathbf{1}\{A_i = 1\},$$

so $\tau_i = \tilde{T}_i$ is the follow-up time actually stored and Y_i the event indicator stored beside it. For a censored episode the data assert $T_i > L_i$ and nothing more: the eventual T_i is unobserved, and it is *not* the case that $T_i = L_i$. Writing the follow-up limit as L_i rather than the conventional C_i keeps it clear of the linkage correctness indicator of §2.1.

$Y_i = 0$ means no accepted observable successor was found before the cutoff. It does not mean “not renewed”: the contract may have been re-procured through a central purchasing body, below publication thresholds, or under a link the precision-first rule declined to accept.

Kaplan–Meier. The survivor function and its complement are

$$S(t) = P(T > t), \quad F(t) = P(T \leq t) = 1 - S(t),$$

read as “the probability that an episode is still without an observable successor at age t ” and “the probability that one has appeared by age t ”. Both are estimated non-parametrically by [Kaplan and Meier \(1958\)](#)’s product-limit estimator

$$\hat{S}(t) = \prod_{t_k \leq t} \left(1 - \frac{d_k}{n_k}\right),$$

where d_k is the number of observable-successor events at event time t_k and n_k the number of episodes still at risk immediately before it. At each observed event time the estimator multiplies in the probability of getting past that instant among the episodes still under observation, which is what lets censored episodes contribute their exposure up to the moment they leave. It is reported overall and stratified by CPV segment, the only stratification this study estimates; group differences use a multivariate log-rank test across those segments, whose hypotheses are

$$\begin{aligned} H_0 : S_{32}(t) &= S_{35}(t) = S_{48}(t) = S_{72}(t) \quad \text{over the observation window,} \\ H_1 : &\text{at least one segment survivor function differs.} \end{aligned}$$

This is an omnibus test: rejecting H_0 says the four curves are not all equal, not that every pair of segments differs. Region and framework status enter the Cox model as covariates but are not estimated as separate Kaplan–Meier strata.

Hazard and the Cox model. The hazard is the conditional event rate among episodes that have survived to t :

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}.$$

It answers “among episodes that have reached age t with no observable successor, how fast are successors appearing right now?” A hazard is a rate, not a probability: it is not bounded by 1, and a hazard ratio is consequently not a risk ratio. The semi-parametric model is

$$\lambda(t \mid X) = \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p), \quad \text{HR}_k = \exp(\beta_k),$$

which leaves the baseline $\lambda_0(t)$ unspecified and reads each covariate as a multiplicative shift of the hazard. $\text{HR}_k > 1$ means a higher instantaneous observable-successor hazard at every t , holding the other included covariates fixed; $\text{HR}_k < 1$ a lower one. Covariates are selected for

substantive relevance and data quality rather than automated search: CPV digital segment, buyer region, framework-agreement flag, validated-SIREN availability, and centered award year. The rule of one covariate per ten observed events (Van Belle et al., 2002) is respected with 544 events supporting 8 covariates.

Proportional-hazards diagnostic. The model above assumes $\text{HR}_k(t) = \exp(\beta_k)$ does not vary with t . The Schoenfeld-residual test (Grambsch and Therneau, 1994) takes

$$H_0 : \text{the hazard ratio for covariate } k \text{ is constant in } t,$$

and a $p < 0.05$ is evidence against it. Where it fails, the coefficient is still a well-defined summary – a time-averaged association – and is reported as such rather than silently dropped or used to discard the model.

Parametric models. Exponential, Weibull, log-logistic, log-normal, and generalized-gamma models are compared by log-likelihood ℓ and the penalised criteria

$$\text{AIC} = 2k - 2\ell, \quad \text{BIC} = k \log n - 2\ell,$$

with k the number of fitted parameters and n the number of episodes. Lower is better *within the compared set*: these criteria rank the five families against one another on a fit-versus-complexity trade-off and say nothing about whether the winner fits in absolute terms, which is why the choice is also checked graphically against the Kaplan–Meier curve. Their role is to identify the best-fitting family and to provide the instrument any extrapolation past the observation window would require. They are *not* the source of the reported 12/24-month probabilities: every horizon quoted in this report falls inside the observed window, and the smooth families flatten the empirical renewal shoulder, so the operational conditional probabilities are read off the Kaplan–Meier estimator, which imposes no shape.

Temporal validation. The model is fit once on episodes awarded 2015–2021 and scored out of time without refitting. The primary evaluation window is 2022–2024, as specified by the internship guideline; 2022–2025 is carried as a sensitivity read that adds the shortest-follow-up award cohort. Harrell’s concordance index is a probability about *pairs*, not a classification accuracy:

$$C \approx P(\text{the model ranks the earlier-event episode as higher risk} \mid \text{the pair is comparable}),$$

where a pair is comparable when censoring allows their orderings to be determined. $C = 0.5$ is ordering no better than chance and $C = 1$ is perfect ordering. It is reported on each split to assess discrimination and out-of-time stability, not to target a specific value.

Interval estimation. The operational conditional probabilities are ratios of two points on the same fitted Kaplan–Meier curve, so their sampling error is not the sum of two independent errors. For a statistic $\hat{\theta}$ the pipeline therefore resamples episodes with replacement, refits Kaplan–Meier on each resample, and reports the empirical percentiles of the replicates $\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(B)}$ with $B = 500$. Resampling whole episodes carries the dependence between $S(a)$ and $S(a + h)$ through without a delta-method derivation or further distributional assumptions.

Evaluation strategy. The three checks applied to the linkage – scoring it against a reference subset, comparing linked with unlinked episodes, and re-running the analysis under alternative linkage rules – follow the guidance of [Harron et al. \(2017\)](#) for evaluating linkage quality in linked-data analyses. That source supports the strategy; it does not validate the precision, recall, or threshold reported here.

Borderline-link robustness. Threshold sensitivity swaps one event definition for another; a separate check asks how much rests on the anchors the frozen rule nearly classified the other way. Anchors whose best candidate scores within ± 0.05 of the 0.70 acceptance bar are dropped entirely – borderline acceptances and borderline abstentions alike – and the headline Kaplan–Meier levels and hazard ratios are recomputed on the remainder. The band is fixed a priori and is not searched over.

Linkage sensitivity. Because the event is linkage-conditioned, the same Kaplan–Meier and Cox analyses are repeated under the strict ($M_B@0.80$), main ($M_B@0.70$), looser ($M_B@0.60$), and high-recall contrast ($M_C@0.70$) event definitions. Writing $\mathcal{L}_m$ for linkage definition m makes the dependence explicit: every survival quantity in this report is really

$$P_m(T \leq t) \quad \text{and} \quad \text{HR}_{k,m},$$

that is, a probability and a hazard ratio computed *under* $\mathcal{L}_m$. The subscript is dropped elsewhere only because $\mathcal{L}_{\text{main}}$ is fixed. The arms are deterministic pre-specified scenarios, not draws from a distribution over linkage rules, so varying m bounds a sensitivity rather than estimating an uncertainty. A conclusion is reported as robust only when it is stable in sign and approximate magnitude across these arms; the recurring finding is that $P_m(T \leq t)$ moves a great deal with m while the sign and rough size of the leading $\text{HR}_{k,m}$ do not.

7 Trend And Change-Point Detection

Quarterly awarded-episode counts $N_{s,q}$ are built for the overall cohort and each CPV digital segment from 2015Q2 (the first complete quarter) through 2025Q4, including zero-count quarters.

These are time-series methods and are stated in their natural forms; forcing them into conditional-probability notation would obscure rather than clarify. The one exception is the hidden Markov model, whose output genuinely is a conditional probability and is written as one below. Greek letters in this section are local to it: the penalty multiplier λ is unrelated to the hazard $\lambda(t)$ of §6.

Change-point detection (PELT). Killick, Fearnhead and Eckley (2012)’s PELT algorithm minimizes, over the number of change points K and their positions $\tau_1 < \dots < \tau_K$,

$$\sum_{k=0}^K \mathcal{C}(y_{\tau_k+1:\tau_{k+1}}) + K\beta, \quad \beta = \lambda \log(n),$$

on the z-standardized series, where $\mathcal{C}$ is within-segment squared error. The first term rewards fitting each segment well; the second charges a fixed price β per break, which is what stops the optimum from putting a break between every pair of quarters. The central result uses $\lambda = 1$ with sensitivity at 0.5 and 2.0, and a break is called stable only if it lies within one quarter under all three multipliers. PELT answers *when* the level of the series shifted; it never answers *why*.

Stationarity. Augmented Dickey–Fuller and KPSS are run on each segment’s series with deliberately opposite nulls,

$$H_0^{\text{ADF}} : \text{the series has a unit root (non-stationary)}, \quad H_0^{\text{KPSS}} : \text{the series is level-stationary},$$

so they are read jointly. Rejecting the ADF null while failing to reject the KPSS null is coherent evidence of stationarity; when the two disagree the correct report is ambiguity over this short window, not a forced binary label.

Regime detection (HMM). A 3-state Gaussian hidden Markov model is fit on the quarter-over-quarter *change* $\Delta N_t = N_t - N_{t-1}$ – not the level – for the overall cohort and the two highest-volume CPV segments. The hidden state $Z_t \in \{\text{decline}, \text{plateau}, \text{growth}\}$ is governed by transition probabilities $P(Z_t = k \mid Z_{t-1} = l)$, and the reported quantity is the filtered posterior

$$P(Z_t = k \mid \Delta N_1, \dots, \Delta N_t),$$

read as: given the observed sequence of quarterly changes and the fitted model, how probable is regime k in the current quarter? States therefore describe typical period-over-period direction rather than absolute activity level. This probability is conditional on the fitted model, not an observed property of the market: a high posterior on **growth** says the model finds that regime most consistent with recent changes, not that the market is demonstrably growing. It complements PELT, which finds discrete historical breaks, with a continuously updated regime read; the two are not required to agree, and disagreement is reported honestly rather than reconciled.

Recent direction. The recent direction comes from an ordinary least-squares fit over the latest 12 quarters,

$$N_t = \alpha + \beta t + \varepsilon_t,$$

where $\hat{\beta}$ is the estimated change in awarded episodes per quarter over that window. A segment is labelled **increasing** or **decreasing** only when $\hat{\beta}$'s two-sided p -value falls below the pre-declared exploratory $\alpha = 0.10$, uncorrected for multiple testing; otherwise it is **stable_or_uncertain**. $\hat{\beta}$ describes the last 12 quarters. It is not a forecast, and no value of N_t beyond the window is implied. None of these methods forecasts future values or identifies the cause of a break; causal attribution requires documentary or stakeholder evidence not available in BOAMP alone. Monetary and duration trend series are omitted because no canonical awarded-amount field is validated at episode grain, and 2025's duration-field completeness jump is a measurement change, not a genuine shift in contract durations.

8 Technology Segmentation

The internship guide's L2 deliverable specifies a supervised technology taxonomy: 300–500 manually annotated contracts across 8–12 classes, a TF-IDF+logistic-regression/SVM baseline evaluated by macro-F1 and confusion matrix, and an optional CamemBERT comparison. It is implemented here as an enrichment layer over the CPV-based study, and reported in full in `TECHNOLOGY_TAXONOMY_REPORT.md`.

The corpus is 500 manually annotated BOAMP notices spanning 2015–2025, labelled into eight substantive classes – cloud/hosting, cybersecurity, network/telecom, IT infrastructure, business software, data/Bi, AI, and IT services – plus three fallback classes for mixed, other-digital, and non-technology procurements. The input is the notice object text alone, represented as TF-IDF word n -grams; buyer identity, geography, dates, amounts, procedure type, and every linkage variable are excluded by design, as is CPV, which serves as the benchmark rather than as a feature. The n -gram range is part of the *search space*, not of the final model: unigrams and unigrams-plus-bigrams were both offered to the grouped inner cross-validation, and every fold selected unigrams alone, which is the representation reported below.

Evaluation is a 3-fold group-aware cross-validation in which every family of related notices – notices sharing a reconstructed episode, and notices whose objects are near-identical under a character analyser – is confined to one fold. Without the second rule, 29 near-duplicate pairs sitting in different episodes would have been split across folds and every reported figure would be optimistic. Hyperparameters are selected by a grouped inner cross-validation inside each training fold.

The selected specification, TF-IDF word unigrams with a class-weighted multinomial logistic regression, reaches an out-of-fold macro-F1 of 0.744 against 0.473 for the best administrative benchmark built from CPV codes and BOAMP descriptors, both searched over the same regularisation range. Uncertainty is estimated by resampling procurement families rather than notices, since notices within a family are near-copies: the paired difference is 0.271 with a 95 result justifying the layer: the business technology class is present in procurement text and is not recoverable from the official classification codes. The high-volume classes are separated reliably; AI, with seven annotated notices, is reported as a rare-class limitation rather than as a measured capability. A CamemBERT comparison was gated on the classical model being materially inadequate, defined in advance as macro-F1 below 0.55; that condition was not met and no transformer was run. The deployed confidence score is the raw class probability: Platt scaling was evaluated against a pre-specified adoption rule and rejected, because its macro-F1 cost exceeded the budget the rule allows.

Two limitations are structural rather than incidental. The corpus was delivered as a single labelled file with no annotator identifier and no second pass, so no Cohen’s kappa can be computed and label reliability cannot be quantified; the guide’s two-annotator design is not met. And the corpus is quota-stratified by class and year, so its class proportions are a property of the annotation design and are not estimates of prevalence.

CPV divisions 32 (telecommunications equipment), 35 (security), 48 (software), and 72 (IT services) therefore remain the definition of the study cohort, the Cox covariate, and the quarterly trend series throughout this report. They are official, zero-missingness EU categories under Regulation 213/2008 and they reproduce exactly. The technology taxonomy sits beside them: mean CPV-segment purity against the predicted taxonomy is 0.3385, so each CPV division holds several business technologies, and the two segmentations are complements rather than substitutes.

Downstream use of the predicted labels is gated twice. A class enters the technology-level survival or trend enrichment only if the classifier separates it out of sample (F1 ≥ 0.65 on at least 10 annotated notices) and it is a substantive technology rather than one of the mixed, other-digital or other residuals, and it carries enough episodes and observed events. The second condition is the familiar one; the first matters more than it looks. Running the technology log-rank test over every class with adequate statistical support, residual buckets included, gives $p = 0.0001$; restricting it to the five substantive classes the classifier actually separates gives $p = 0.036$. The residual buckets have distinctive timing because of what they contain, so including them manufactures part of an apparent technology effect.

9 Pilot Reference Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	n/a	3	0.667	0.400	0.091	0.188
M_B_text_ranking	70.0	2	0.500	0.200	0.000	0.125
M_C_weighted_gated	70.0	5	0.800	0.800	0.000	0.312
M_D_fellegi_sunter	65.0	2	0.000	0.000	0.091	0.125

Table 3: Anchor-level metrics on the pilot split, on which the threshold was frozen. The pilot carries 5 reviewed-positive anchors, so these rates are read for direction only.

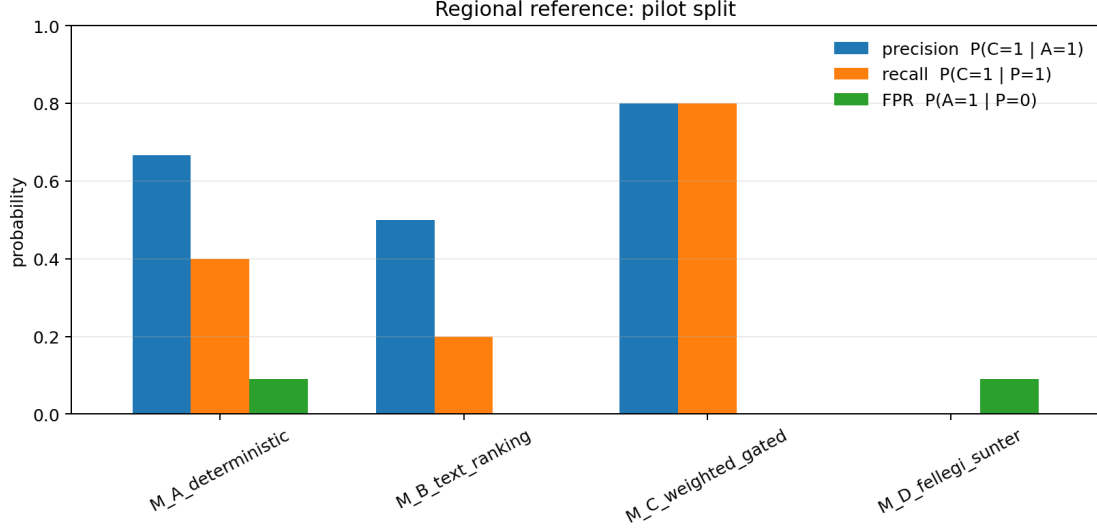


Figure 1: The pilot split as conditional probabilities. It is shown for completeness; the held-out reading is the locked split below.

10 Locked Reference Results

The table below is an internal method-comparison diagnostic. It must not be presented as external validation.

What is being scored. The decision is not a yes/no call on a pre-formed pair. For each anchor i , the reference names a successor identity $R_i \in J_i \cup \{\emptyset\}$ and the method returns an identity $\hat{R}_i \in J_i \cup \{\emptyset\}$, so an anchor with a reviewed successor can still be scored wrong by accepting the *wrong* candidate. That is why the three indicators of §2.1 are needed rather than a single “correct/incorrect” bit:

$$P_i = \mathbf{1}\{R_i \neq \emptyset\}, \quad A_i = \mathbf{1}\{\hat{R}_i \neq \emptyset\}, \quad C_i = \mathbf{1}\{\hat{R}_i = R_i \neq \emptyset\}.$$

The familiar counts follow from them: $TP = \sum_i C_i$; a false positive is either $A_i = 1, P_i = 0$ (accepted where the reference has nothing) or $A_i = 1, P_i = 1, C_i = 0$ (accepted the wrong candidate); a false negative is either an abstention on a positive anchor or that same wrong acceptance. Note that a wrong acceptance is counted on both sides, which is exactly right and is invisible in the $TP/(TP + FP)$ shorthand.

The four metrics as conditional probabilities. Each answers a different question, distinguished by what is conditioned on:

$$\underbrace{P(C_i = 1 \mid A_i = 1)}_{\text{precision}}, \quad \underbrace{P(C_i = 1 \mid P_i = 1)}_{\text{recall}}, \quad \underbrace{P(A_i = 1 \mid P_i = 0)}_{\text{false-positive rate}}, \quad \underbrace{P(A_i = 1)}_{\text{coverage}}.$$

Precision and recall are the *same event under reversed conditioning*, and this is the single most useful thing for a reader to hold on to. Precision asks: given that the method committed to a successor, how often was it the reviewed one? Recall asks: given that a reviewed successor exists, how often did the method recover it? $P(C = 1 \mid P = 1)$ and $P(C = 1 \mid A = 1)$ are not interchangeable and need not be close, because the conditioning sets – 18 reviewed-positive anchors and 8 accepted links – are different populations. The false-positive rate conditions on the *third* population, the 54 anchors the reference found nothing for, so it is not $1 - \text{precision}$; the two share no denominator.

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	n/a	15	0.533	0.444	0.130	0.208
M_B_text_ranking	70.0	8	0.875	0.389	0.000	0.111
M_C_weighted_gated	70.0	23	0.522	0.667	0.185	0.319
M_D_fellegi_sunter	65.0	5	0.200	0.056	0.018	0.069

Table 4: Anchor-level linkage performance on the locked split of the regional reference. Precision $P(C = 1 \mid A = 1)$ is the probability that an accepted successor is exactly the reviewed one; recall $P(C = 1 \mid P = 1)$ the probability of recovering the reviewed successor when one exists; FPR $P(A = 1 \mid P = 0)$ the probability of accepting anything where the reference found nothing; coverage $P(A = 1)$ the share of anchors that receive a link rather than an abstention. Unweighted sample estimates on 72 anchors.

The frozen rule, read cell by cell. On the locked split `M_B_text_ranking` @ 0.70 accepts 8 successors across 72 anchors, of which 7 are the reviewed successor and 1 is a wrong candidate on an anchor that does have one. It abstains on 10 reviewed-positive anchors and on all 54 reviewed-negative anchors. Hence

$$\hat{P}(C = 1 \mid A = 1) = \frac{7}{8} = 0.875, \quad \hat{P}(C = 1 \mid P = 1) = \frac{7}{18} = 0.389,$$

$$\hat{P}(A = 1 \mid P = 0) = \frac{0}{54} = 0.000, \quad \hat{P}(A = 1) = \frac{8}{72} = 0.111.$$

Read them as follows. Among the 8 links the method committed to, 7 matched the reviewed successor. Of the 18 anchors the reference says have a successor, the method recovered 7; the 11 it did not are 10 abstentions and 1 wrong acceptance, and both count identically against recall even though only the second is a fabricated link.

Four caveats travel with these four numbers and none of them is optional. The precision estimate rests on 8 accepted links, so its 95% interval is [0.529, 0.978]: one changed decision moves it materially. Recall’s ceiling is not 1 but 0.913, because $P(C = 1 \mid P = 1) \leq P(E = 1 \mid P = 1)$ – a successor that blocking discarded cannot be recovered by any scorer. An FPR of 0.000 is a diagnostic on 54 corpus-relative negatives on this small sample, not evidence that the population false-positive rate is zero. And coverage is deliberately low: it is a consequence of prioritising $P(C = 1 \mid A = 1)$, not a target to raise, because in a survival dataset a false link fabricates both an event and an event time whereas an abstention is handled honestly as right-censoring.

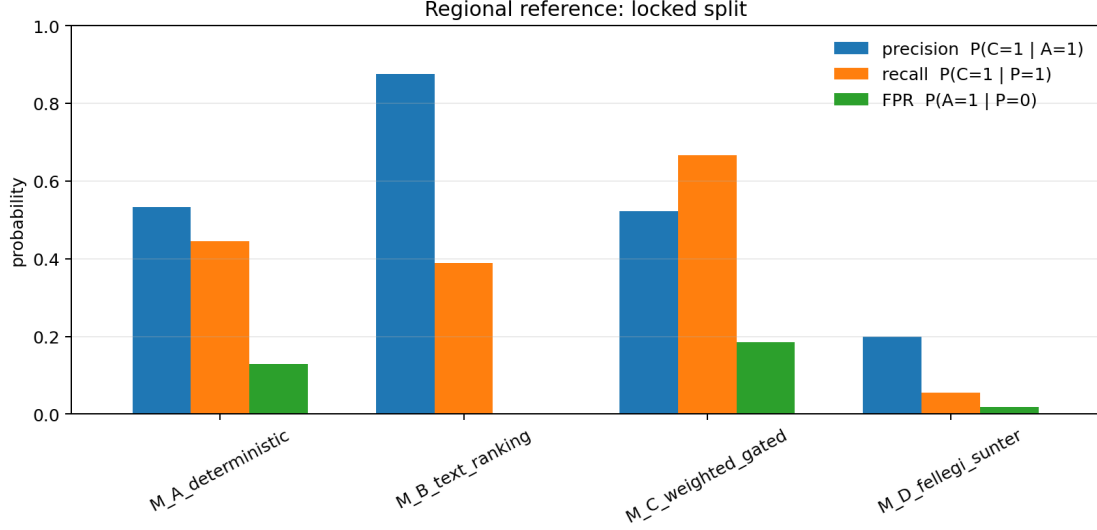


Figure 2: The same three conditional probabilities per method. M_C buys recall $P(C = 1 | P = 1)$ with a materially higher $P(A = 1 | P = 0)$; M_B takes the opposite trade, which is the one this study needs.

11 Quality Evidence And Interpretation

The held-out result supports retaining a conservative baseline for continued work; it does not establish final accuracy. Within this reference the direction is coherent, with M_B best on $P(C = 1 | A = 1)$ and lowest on $P(A = 1 | P = 0)$ among useful methods and M_C showing the expected trade-off, but the intervals in §10 overlap heavily and must be read before separating any two methods.

The pair-level ROC and precision-recall curves should be read as score-ranking diagnostics, not as the final operating decision. The final pipeline decision is anchor-level: choose one successor or abstain. The ROC curve is stepped rather than smooth because the held-out reference contains a finite number of labelled pairs and many tied or near-tied scores.

The emphasis on precision-recall for rare positive decisions is supported by [Davis and Goadrich \(2006\)](#) and [Saito and Rehmsmeier \(2015\)](#). These sources justify the diagnostic choice; they do not validate the reference labels, numerical results, or selected threshold. Generic web illustrations are therefore treated as presentation aids only, not as academic evidence.

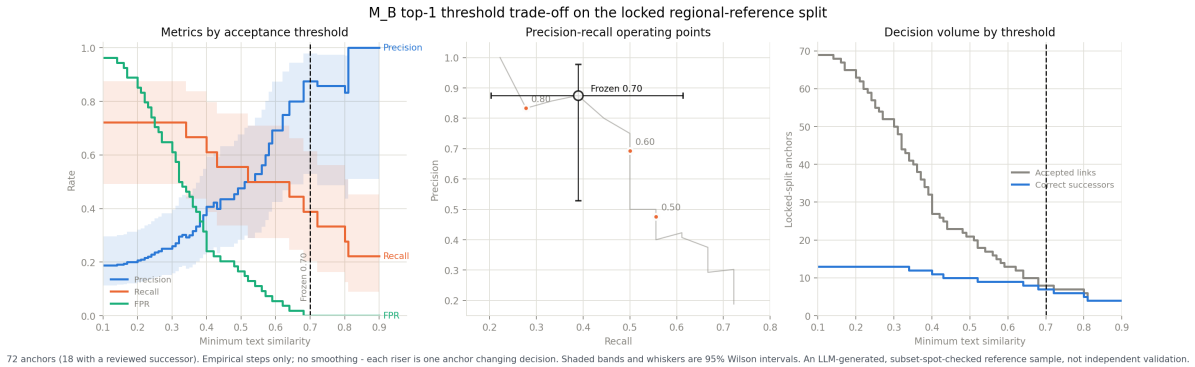


Figure 3: Project-specific, unsmoothed anchor-level threshold trade-off for M_B . The figure is empirical evidence from the locked split of the regional reference, not an idealised illustration.

On the locked split, 0.70 accepts 8 links of which 7 are correct (precision 0.875, recall 0.389,

FPR 0.000), while 0.60 accepts 13 of which 9 are correct (precision 0.692, recall 0.500, FPR 0.056). On the pilot split the same two points give precision 0.500 and 0.667 respectively. The threshold was fixed before this reference was read, which is the only reason the locked split can be reported as held out; selecting one from these rows now would convert it into a tuning set. The lower threshold is therefore reported as a sensitivity arm rather than promoted after inspecting validation.

12 Model-Assisted Challenge Review

A separate blinded challenge review sampled 20 accepted links, 20 high-similarity structural-negative candidates, and 20 buyer-declared relationships. Its provenance is model-assisted rather than independent human specialist review. Among the 20 accepted links, 14 were confirmed, 5 were rejected, and 1 was uncertain. Conservative reviewed precision was 0.700, with exact 95% interval [0.457, 0.881]. This diagnostic falls below the 0.80 point target and exposes useful failure modes. It does not establish independent human validation, but it is sufficient to justify keeping the reported claim narrow and the lower threshold unpromoted.

13 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

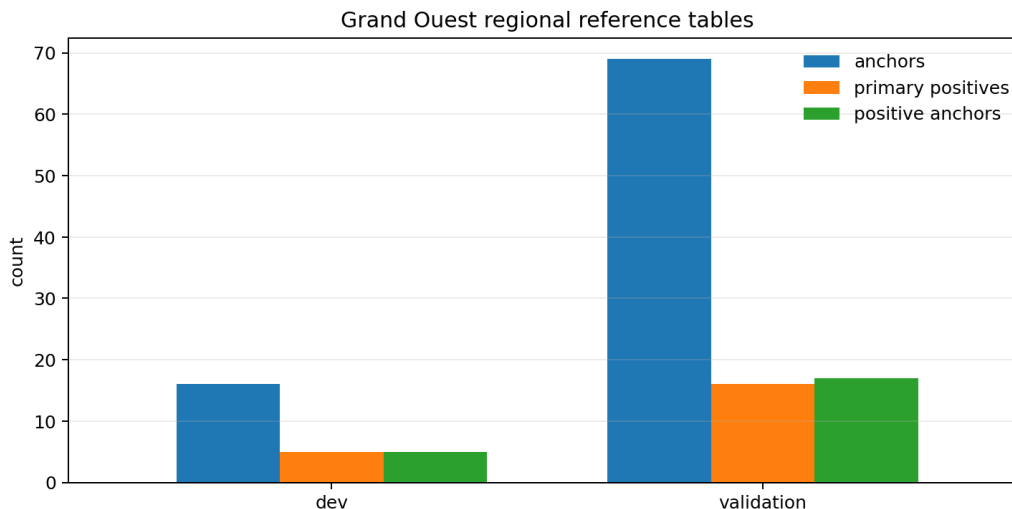


Figure 4: Row and label counts behind each modelling table, so a reader can see how thin the positive support is before reading any metric computed on it.

14 Survival Results

The survival dataset uses the main accepted links as events and all other episodes as right-censored observations at the study cutoff. The main variant contains 544 events and 3,256 censored observations (14.316% event rate). Segment event rates are 0.132 for CPV-32, 0.204 for CPV-35, 0.117 for CPV-48, and 0.143 for CPV-72. CPV-35 has the highest observed successor rate in the main arm.

Kaplan–Meier. The cumulative event probability $F(t) = 1 - S(t)$ is estimated at

$$\hat{F}(12) = 0.04621, \quad \hat{F}(24) = 0.06733, \quad \hat{F}(60) = 0.17535,$$

that is 4.621% by 12 months, 6.733% by 24 months and 17.535% by 60 months. These are linkage-conditioned probabilities that an observable successor procurement becomes visible in BOAMP for an episode of the 3,800-episode study cohort under $M_B@0.70$. They are not certified renewal probabilities, and because missed successors push the level down while residual false links push it up, they are not one-sided bounds either.

The Kaplan–Meier median survival time is **not reached**: $\hat{S}(t)$ never falls below 0.5 within the observation window, so no median is reported. This is distinct from the 31.82-month median delay *among linked events only*, which conditions on the event having occurred. A multivariate log-rank test of $H_0 : S_{32} = S_{35} = S_{48} = S_{72}$ across CPV segments gives statistic 23.45 ($p = 3.26 \times 10^{-5}$), so the data reject equality of the four segment curves. Being an omnibus test, it does not establish that every pair of segments differs.

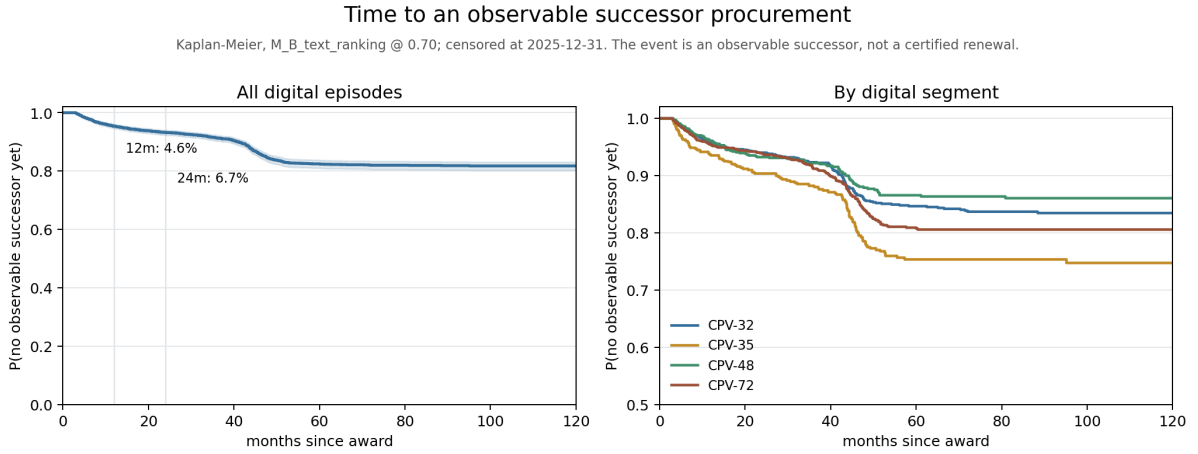


Figure 5: Estimated $\hat{S}(t) = \hat{P}(T > t)$, the probability that an episode is still without an observable successor at age t . The left panel carries the absolute level, which is linkage-conditioned and quoted only with that caveat; the right panel carries the segment ordering, which survives every sensitivity arm.

Cox model. The parsimonious model (8 covariates, 544 events) gives:

Covariate	HR	95% CI	p
framework_flag	1.751	[1.435, 2.136]	3.38e-08
has_validated_siren	1.082	[0.885, 1.323]	0.443
award_year_centered	1.107	[1.066, 1.149]	1.21e-07
digital_segment_CPV-35	1.553	[1.218, 1.981]	0.000383
digital_segment_CPV-48	0.828	[0.638, 1.073]	0.153
digital_segment_CPV-72	1.056	[0.850, 1.310]	0.624
buyer_region_Normandie	0.800	[0.640, 1.000]	0.0501
buyer_region_Pays de la Loire	1.003	[0.815, 1.234]	0.979

Table 5: Cox hazard ratios $HR_k = \exp(\beta_k)$ for the observable-successor hazard, main linkage arm. $HR_k > 1$ means the covariate is associated with a successor appearing sooner. In-sample C-index 0.626.

Framework-agreement episodes and CPV-35 carry the largest hazard ratios among substantively interpretable covariates. The framework coefficient reads: conditional on an episode still

having no observable successor at time t , and holding the other included covariates fixed, framework episodes have an estimated hazard 1.751 times that of non-framework episodes – roughly 75.1% higher. CPV-35 reads the same way at 1.553 against the CPV-32 reference segment. Three qualifications apply to both. These are descriptive associations with the observed hazard, not causal effects. A hazard ratio is not a risk ratio, so neither number says an episode is 75.1% more likely to be renewed. And the proportional-hazards assumption is rejected for `framework_flag` below, so its ratio is a time-averaged association rather than a constant multiplier.

Covariate	Test statistic	p	PH assumption
<code>award_year_centered</code>	70.733	4.09e-17	violated
<code>buyer_region_Normandie</code>	0.608	0.436	ok
<code>buyer_region_Pays de la Loire</code>	2.455	0.117	ok
<code>digital_segment_CPV-35</code>	0.041	0.84	ok
<code>digital_segment_CPV-48</code>	0.671	0.413	ok
<code>digital_segment_CPV-72</code>	3.723	0.0537	ok
<code>framework_flag</code>	7.040	0.00797	violated
<code>has_validated_siren</code>	6.763	0.00931	violated

Table 6: Schoenfeld-residual tests of H_0 : the hazard ratio for this covariate is constant over time. A small p is evidence against proportionality, not evidence that the covariate does not matter.

Proportional-hazards diagnostic. The assumption is rejected ($p < 0.05$) for `award_year_centered`, `framework_flag`, `has_validated_siren`. These coefficients are therefore reported as time-averaged associations rather than constant hazard ratios; stratification or time-interaction terms would be the next refinement if individualized prediction were required. Nothing in the operational deliverable depends on proportionality, because the 12/24-month probabilities come from the Kaplan–Meier estimator, which makes no such assumption.

Temporal validation. Training on 2015-2021 (2,470 episodes, 392 events) and evaluating on the guideline-aligned 2022-2024 window (1,004 episodes, 107 events) gives C-index 0.606 in-sample versus 0.479 out-of-time. Extending the test window to 2022-2025 (1,330 episodes, 152 events), without refitting, gives 0.518. Read these as pairwise ranking probabilities: among comparable pairs of episodes drawn from the test window, the model puts the one that gets its successor first at higher risk about 47.9% of the time, where 50% is chance. This is *not* a classification accuracy of 47.9%; no episode is being classified. Both figures sit close to the 0.5 chance line, so individualized out-of-time discrimination is weak. The two windows also differ in follow-up, and [Uno et al. \(2011\)](#) show that the concordance statistic for right-censored data converges to a quantity depending on the censoring distribution, so the gap between the two figures should not be read as a change in discriminative ability. That is a result, not a prompt to retune – the model is retained as a descriptive risk-factor summary and nothing in the operational deliverable rests on it. Part of the gap is structural, because episodes awarded from 2022 onwards can contribute only short-gap events.

Borderline-link robustness. Dropping the 280 episodes whose best candidate scores within ± 0.05 of the acceptance threshold removes 133 events and leaves the direction of both headline hazard ratios unchanged (CPV-35 1.55 to 1.78; framework 1.75 to 1.62). The absolute Kaplan–Meier level falls, from 4.6% to 3.7% at 12 months, which is the mechanical consequence of removing borderline events. The comparative conclusions therefore do not rest on borderline linkage decisions, while absolute probabilities remain threshold-uncertain.

Model	Parameters	Log-likelihood	AIC	BIC
GeneralizedGamma	3	-3757.5	7521.0	7539.7
LogNormal	2	-3784.6	7573.2	7585.7
LogLogistic	2	-3807.3	7618.6	7631.1
Weibull	2	-3814.7	7633.4	7645.8
Exponential	1	-3829.7	7661.3	7667.6

Table 7: Parametric survival model comparison by $AIC = 2k - 2\ell$ and $BIC = k \log n - 2\ell$. Lower is better *among these five families*; neither criterion establishes that the winner fits the data.

Parametric models. `GeneralizedGamma` has the lowest AIC and BIC and was checked graphically against the Kaplan–Meier curve. It is reported as the best-fitting family and as the instrument any extrapolation past the 2025-12-31 cutoff would use. It is *not* the source of the exported 12/24-month conditional successor probabilities (`survival_conditional_probabilities.csv`): those come from the Kaplan–Meier estimator, each with a 500-draw episode-bootstrap interval. The reason is visible in the fit itself – around age 36 months, just before the observed renewal shoulder, every parametric family including the selected one reports roughly 0.02–0.03 for the next twelve months where Kaplan–Meier reports near 0.075. Since every horizon this report quotes falls inside the observed window, the extrapolation that would justify accepting that smoothing is not needed. These are estimated probabilities of an *identifiable observable successor procurement*, not certified renewal probabilities, and no active Gigalis portfolio was available to score, so they cover the study cohort rather than a live deployment set.

Operational 12- and 24-month probabilities. This is the quantity the business question actually asks for, and it is the clearest mathematical statement of the internship’s operational problem. For an episode that has reached age a months with no accepted observable successor, the probability that one becomes visible within the next h months is

$$P(T \leq a + h \mid T > a) = P(a < T \leq a + h \mid T > a) = 1 - \frac{S(a + h)}{S(a)},$$

the two left-hand forms being identical because the conditioning event $\{T > a\}$ already excludes $T \leq a$. Here a is the episode’s current age, h the forward horizon, and $\{T > a\}$ the statement that no observable successor has appeared yet. The quantity is read off the Kaplan–Meier estimator with the 500-draw episode-bootstrap intervals described in §6.

Age a at assessment	$P(T \leq a+12 \mid T > a)$	95% CI	$P(T \leq a+24 \mid T > a)$	95% CI
0 months	4.62%	[3.91, 5.24]	6.73%	[5.91, 7.58]
12 months	2.21%	[1.70, 2.73]	4.26%	[3.55, 4.98]
24 months	2.09%	[1.58, 2.60]	9.40%	[8.33, 10.68]
36 months	7.46%	[6.52, 8.59]	9.69%	[8.58, 11.12]
48 months	2.41%	[1.76, 3.07]	2.89%	[2.15, 3.72]

Table 8: Given that an episode has reached age a with no accepted observable successor, the estimated probability that one becomes visible in the following 12 or 24 months. Kaplan–Meier on the 3,800-episode cohort under $M_B@0.70$.

Worked example, the row a purchasing body would use most. At $a = 36$ months and $h = 12$,

$$\hat{P}(T \leq 48 \mid T > 36) = 0.0746, \quad 95\% \text{ CI } [0.0652, 0.0859].$$

In words: conditional on an episode reaching 36 months with no accepted observable successor, the estimated probability that one becomes visible in the following 12 months is 7.46%.

Probability that a successor becomes visible in the next 12 or 24 months

Kaplan-Meier conditional probabilities with 95% episode-bootstrap intervals; the profile peaks into the 36-48 month renewal shoulder.

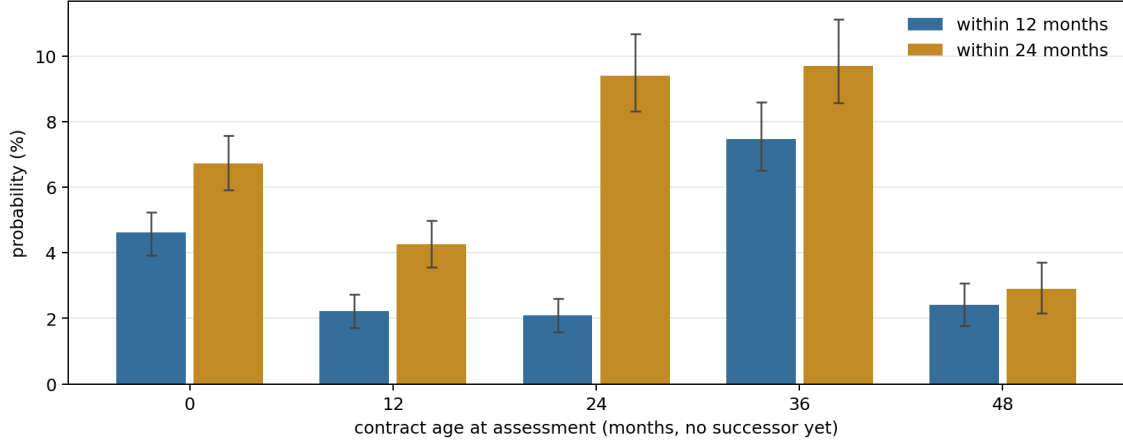


Figure 6: The same quantities with their bootstrap intervals. The profile is not monotone in a : it rises into the 36–48 month renewal shoulder and falls away after it, which is the empirical feature every parametric family smooths out.

The intervals are wide relative to the estimates, and these numbers rank ages and segments rather than calibrating individual forecasts. They estimate an observable successor procurement appearing in BOAMP, not a certified renewal, and they are conditional on the linkage rule.

Template-risk robustness. The four linkage arms and the borderline band both perturb where the acceptance bar sits. Neither addresses the false-positive mechanism identified above, because that mechanism produces links well *above* the bar: shared framework boilerplate can drive the character analyser high between unrelated objects, and because M_B ranks candidates within each anchor independently, one such episode can be accepted for several anchors. A stricter threshold does not remove either signature and can enrich for them. Two observable signatures, both published by the candidate-generation audit, define a risk flag on each accepted link:

$$Q_i = \mathbf{1}\{\text{word-level similarity} < 0.50 \text{ or } \text{successor reused across anchors}\},$$

carrying 65 and 127 links respectively and 173 in union, so that

$$\hat{P}(Q = 1 \mid A = 1) = \frac{173}{544} = 0.318.$$

This is the share of accepted production links carrying a conservative risk signature. It is emphatically not $P(\text{false link} \mid A = 1) = 0.318$. A flag is a reason to stress-test a link, not a finding that it is wrong; inspection says most flagged links are legitimate rewordings and multi-lot programmes. Those anchors are re-censored at the cutoff rather than dropped, because if a link were spurious the anchor had no observed successor and should still contribute its full follow-up as censored exposure.

Analysis	Episodes	Events	KM 12m	KM 24m	CPV-35 HR	Framework HR
Main	3,800	544	4.62%	6.73%	1.553	1.751
Re-censoring template risk	3,800	371	2.64%	3.88%	1.541	1.692

Table 9: Headline results with the $Q_i = 1$ links re-censored. The absolute event level moves strongly; the relative associations barely move. That contrast, not either row on its own, is the finding.

This is the check the framework-agreement finding most needs, since framework boilerplate is the text driving the mechanism: were the higher framework hazard an artefact of shared legal wording, re-censoring would collapse it. It does not. Events fall from 544 to 371 and $\hat{F}(12)$ from 4.62% to 2.64%, while CPV-35 moves 1.553 to 1.541 and framework 1.751 to 1.692. Both hazard ratios keep their side of 1, so the comparative findings are not products of the documented false-positive mechanism. The fall in the absolute Kaplan–Meier level is roughly the share of events re-censored, which is arithmetic rather than evidence. The check *bounds* how much the mechanism could be moving the results; it does not establish that the flagged links are false, and most of them are not.

Detectability and selection diagnostic. If links were found mainly where records happen to be well documented, the event set would be a biased sample and the survival estimates would inherit that bias. Linked ($Y_i = 1$) and censored ($Y_i = 0$) episodes are therefore compared on the standardized mean difference

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}},$$

which measures the gap between the two groups in pooled within-group standard deviations and so is comparable across variables on different scales. The largest observed gap is `log_candidate_pool_size` at 0.470. This indicates possible differential detectability – longer notice text gives the text scorer more to match on – not proof of causal linkage bias, and it cannot be fully separated from genuine heterogeneity in renewal behaviour using BOAMP alone.

Linkage sensitivity. Event counts range from 296 to 1,332 across the four retained linkage arms, so $P_m(T \leq t)$ is linkage-sensitive by construction: a looser $\mathcal{L}_m$ manufactures more events. The question the table below answers is whether $\text{HR}_{k,m}$ is equally sensitive.

Covariate	Strict	Main	Looser	High-recall	Robustness
award_year_centered	1.017	1.107	1.136	1.101	Robust
buyer_region_Normandie	0.803	0.800	0.794	0.726	Mostly Robust
buyer_region_Pays de la Loire	0.977	1.003	1.023	0.891	Linkage Sensitive
digital_segment_CPV-35	1.792	1.553	1.488	1.484	Robust
digital_segment_CPV-48	0.652	0.828	0.807	0.885	Mostly Robust
digital_segment_CPV-72	1.108	1.056	1.036	1.557	Linkage Sensitive
framework_flag	1.281	1.751	1.953	1.623	Robust
has_validated_siren	1.193	1.082	1.107	1.650	Linkage Sensitive

Table 10: $\text{HR}_{k,m}$ for each covariate k under each linkage definition $\mathcal{L}_m$. Reading a row across is the robustness test: a covariate whose ratio keeps its side of 1 across a fourfold change in event count is not an artefact of where the acceptance bar was placed.

Framework flag, CPV-35, and centered award year are the most robust associations; buyer-region and CPV-72 effects are linkage-sensitive and should not be over-interpreted. The overall

pattern is the one the notation was introduced to name: $P_m(T \leq t)$ varies considerably with m , while the direction and rough size of the leading $\text{HR}_{k,m}$ do not.

15 Trend Results

Segment	Direction	Slope/qtr	Raw p	Holm p	BH p	Last stable PELT break	HMM regime
Overall	stable_or_uncertain	-0.11	0.921	1.000	0.989	–	growth
CPV-32	stable_or_uncertain	-0.01	0.989	1.000	0.989	2020Q2	growth
CPV-35	stable_or_uncertain	0.03	0.923	1.000	0.989	–	–
CPV-48	decreasing	-0.84	0.032	0.159	0.159	2024Q1	–
CPV-72	stable_or_uncertain	0.70	0.285	1.000	0.714	2021Q1	growth

Table 11: Trend signal matrix: OLS 12-quarter slope, PELT breaks, and HMM current regime (Overall and top-2 segments only).

CPV-48 carries the clearest nominal 12-quarter decline (raw $p = 0.032$), but the five segment slopes in this table are fitted and read simultaneously, and it does not survive that multiplicity: Holm $p = 0.159$, BH $p = 0.159$, against the pre-declared exploratory $\alpha = 0.1$. It is reported as an exploratory signal to monitor, not as an established decline, and the same correction is applied to the technology trend family. The remaining segments are **stable_or_uncertain** before any adjustment.



Figure 7: Quarterly awarded digital procurement episodes, 2015Q2–2025Q4. Dashed lines mark PELT breaks that survive all three penalty multipliers. A break dates a level shift; it does not explain one.

Operational reading. Each segment’s signals translate into a monitoring action, carried in the `business_recommendation` column of `trend_signal_matrix.csv` so that this text and the materialised table cannot drift apart. These are readings of descriptive evidence, not forecasts and not causal explanations: a PELT break dates a level shift without explaining it, and none of the statements below should be quoted as attributing a shift to policy, COVID, regulation, or technology.

- **Overall** (stable_or_uncertain): Maintain monitoring for Overall; no statistically clear recent direction at the pre-declared exploratory level. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.
- **CPV-32** (stable_or_uncertain): Maintain monitoring for CPV-32; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2020Q2; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.
- **CPV-35** (stable_or_uncertain): Maintain monitoring for CPV-35; no statistically clear recent direction at the pre-declared exploratory level.

- **CPV-48** (decreasing): Treat the recent CPV-48 decline as exploratory: it is the clearest nominal signal in the panel, but it does not survive correction for the simultaneous segment tests. Watch it for another few quarters before acting on it. A penalty-stable level shift is dated 2024Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause.
- **CPV-72** (stable_or_uncertain): Maintain monitoring for CPV-72; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2021Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.

Segment	ADF (H0: unit root)	KPSS (H0: level stationary)	Note
Overall	stat=-5.890, p=0.000	stat=0.268, p=0.100	
CPV-32	stat=-6.086, p=0.000	stat=0.808, p=0.010	
CPV-35	stat=-1.684, p=0.439	stat=0.444, p=0.058	
CPV-48	stat=-1.749, p=0.406	stat=0.545, p=0.032	
CPV-72	stat=-5.439, p=0.000	stat=0.611, p=0.022	

Table 12: Stationarity diagnostics per segment.

ADF and KPSS test opposite null hypotheses and are read jointly; disagreement between them means the series is not cleanly classified as stationary or non-stationary over this 43-quarter window.

The HMM regime column above reports each covered segment’s current-quarter regime and posterior probability, fit on the quarter-over-quarter change in episode count rather than its level. Because the model is fit on noisy, low-count quarterly data, the **plateau** state is a data-driven middle tier rather than a change centered exactly at zero, and it is read alongside **decline/growth** rather than as "no change." The HMM’s current-quarter regime and the 12-quarter OLS slope are complementary diagnostics computed over different windows; they are not required to agree, and this report does not adjust either to match the other.

16 Final Linkage Specification

The final event definition is **M_B_text_ranking @ 0.70**, a frozen conservative observable-successor baseline, with the stricter ($M_B@0.80$), looser ($M_B@0.60$), and weighted-gated ($M_C@0.70$) variants as the required sensitivity analyses, together with the borderline-band check. The threshold is not claimed to be optimal; it is claimed to be pre-specified. It was fixed before the regional reference was consulted and has not been moved since, so the locked-split figures are held out rather than fitted. Moreover, the completed model-assisted review confirmed only 14 of 20 sampled production links at 0.70 conservatively, so lowering an unreviewed threshold would not support a stronger accuracy claim. This is a precision-first design. Low recall is accepted as an explicit trade-off, but the observed event rate is not a mathematical lower bound: missed successors push it downward while residual false links can push it upward. [Doidge and Harron \(2019\)](#) make the general point that missed and false links misclassify in opposite directions, which is why no one-sided bound follows from a linkage-conditioned event rate. It is therefore a linkage-conditioned indicator whose absolute value must be read with the strict, looser, and weighted-gated sensitivity results and the borderline-band check. Independent specialist review is required only before claiming externally validated precision or promoting a new threshold, not to complete this linkage-conditioned descriptive study. If that review shows weaker precision, the claim should be narrowed further rather than forcing a more complex model.

Segment continuity of the accepted links. Because hard same-CPV blocking is not imposed, the accepted links are checked after the fact for segment drift. Of the 544 accepted primary links, 538 have a CPV division observed on both sides; 351 (65.2%) stay inside one division and 187 (34.8%) cross divisions, while 75.8% share at least one division across their full CPV lists. The comparison that interprets this is the reviewed reference, which crosses divisions at 39.1% – close to the 34.8% observed among accepted links. Relaxed CPV blocking therefore reproduces roughly the segment-continuity behaviour the reviewed successors already display, rather than introducing drift of its own. The largest cross-division flows are the goods-to-services pairs named above. Full transition counts are in `data/processed/boamp/candidate_generation_cpv_transitions.csv`.

Inspecting the cross-division links by hand also identifies the dominant false-positive mechanism, which is textual rather than categorical. French award notices carry long standardised framework-agreement boilerplate – *accord-cadre mixte*, *bons de commande*, *bordereau des prix unitaires*, *marchés subséquents* – and for buyers whose notices are dominated by that template, character-level similarity can clear the threshold on shared legal phrasing alone. 65 accepted links carry word-level similarity below 0.50 and so rest on the character analyser; inspection shows most are legitimate rewordings of the same object, but this is also the band in which the failure mode lives. Because M_B ranks candidates within each anchor independently, with no one-to-one constraint, one episode can be accepted for several anchors: the most reused successor here is accepted by 11 anchors, and 44 of the 461 distinct successors serve more than one. Multi-lot programmes make much of that legitimate, but the clearest spurious instance found by inspection is a single catering framework accepted as the successor of nine unrelated digital anchors from one regional authority, whose notices are almost entirely template text. A hard same-CPV block would suppress that particular case but would not fix the mechanism, which also operates within a division, and would cost the 9 genuine reviewed successors quantified above. The mechanism is therefore documented as a known contributor to the measured imprecision rather than patched, and it is one reason the absolute event level is reported as linkage-sensitive.

17 Limitations And Robustness

- BOAMP does not consistently encode legal renewal status; accepted links are observable successor procurements, not legal renewal proof.
- The reference labels are a spot-checked LLM research pass, not anchor-by-anchor verification and not an independent specialist panel, and its negatives are corpus-relative, so the reported false-positive rate is a sample diagnostic rather than a population-wide rate (§4).
- The held-out reference is small: 18 positive anchors and 8 accepted M_B links. The selected threshold should not be over-tuned and its accuracy remains provisional.
- Buyer standardisation is improved but remains an important risk area. The legal-form audit is retained to catch name-only and cross-legal-form cases.
- Missing duration and expiry fields are not imputed. This avoids creating false timing certainty.
- The 3-state HMM regime labels are fit on noisy, short (43-quarter) count series; state-count and window choices are not exhaustively validated, and a ‘plateau’ label is a relative middle tier rather than a change centered at zero.
- Cohort selection, Cox covariates, and trend series use CPV divisions; the learned taxonomy (§8) enriches them rather than replacing them, and its labels are predictions carrying a measured error rate, disclosed rather than concealed.

18 Perspectives

18.1 Causal Inference (Outline Only)

The internship guide frames a distinct causal question, separate from this report’s descriptive linkage/survival/trend results: does a Gigalis pooled framework agreement actually change member purchasing behaviour (amounts, frequency, segments), or does Gigalis simply capture demand that would have existed anyway? **This question is not answered here, and no causal estimate is computed in this repository.** Answering it requires data this project’s BOAMP-only corpus does not contain: which buyers are Gigalis members, and when each joined. Attempting a substitute causal claim from BOAMP alone – for example treating CPV-segment membership or region as a proxy treatment – would confound procurement domain and geography with Gigalis adoption, and this project has consistently avoided exactly this kind of unsupported inferential leap elsewhere (survival hazard ratios are reported as associations, not effects, for the same reason).

If Gigalis-internal membership and adoption-date data were joined to this corpus, the natural identification strategy is a **staggered-adoption difference-in-differences** design ([Callaway and Sant’Anna, 2021](#)), because members join the central purchasing body at different calendar dates rather than simultaneously – the specific setting that motivates that estimator over classical two-period DiD. The parallel-trends assumption (comparable purchasing trajectories absent Gigalis membership) would need to be assessed with pre-adoption trend plots per member cohort. Propensity-score matching ([Angrist and Pischke, 2009](#)) is a secondary candidate if adoption is not sharply timed but depends on observable buyer characteristics (size, budget, sector); regression discontinuity would apply only if a hard eligibility threshold (e.g. a population cutoff) determines Gigalis access, which is not currently known to be the case. Any of these designs should state its identifying assumptions as an explicit causal graph ([Pearl, Glymour and Jewell, 2016](#)) before estimation, not after.

This section is a design outline for future work conditional on Gigalis supplying membership data, not a promise that such data will become available within this internship.

19 Interpretation And Scope Of Claims

The technical report should defend the project as a reproducible measurement pipeline, not as a black-box prediction system. The strongest statement is:

Because legal renewal status is not directly observed in BOAMP, the study estimates observable successor procurements. Candidate generation restricts comparison to the same buyer and a plausible future interval; the selected method then accepts only the strongest textually similar candidate above a fixed threshold. This prioritises precision over recall, which is appropriate because false positive links would create artificial survival events.

20 Reproducibility Artifacts

- `data/processed/boamp/linkage_evaluation_dev.json`
- `data/processed/boamp/linkage_evaluation_validation.json`
- `data/processed/boamp/regional_benchmark/modeling/modeling_summary.json`
- `data/processed/boamp/regional_benchmark/regional_benchmark_manifest.json`
- `data/reference/regional_link_benchmark/`

- `data/processed/boamp/survival_dataset_summary.json`
- `data/processed/boamp/linkage_candidates_summary.json`
- `data/processed/boamp/survival_analysis_summary.json`
- `data/processed/boamp/survival_cox_results.csv`
- `data/processed/boamp/survival_ph_diagnostics.csv`
- `data/processed/boamp/survival_parametric_comparison.csv`
- `data/processed/boamp/survival_cox_linkage_sensitivity.csv`
- `data/processed/boamp/trend_analysis_summary.json`
- `data/processed/boamp/trend_signal_matrix.csv`
- `DATA_QUALITY_REPORT.md`
- `TREND_ANALYSIS_REPORT.md`
- `SURVIVAL_ANALYSIS_REPORT.md`
- `PROJECT_WORK_PROTOCOL.md`
- `REVIEW_AUDIT_RESULTS.md`
- `EXECUTIVE_SUMMARY.md`

21 Methodological References

The source and identity definitions follow the official BOAMP and INSEE documentation. CPV interpretation follows Commission Regulation (EC) No 213/2008. TF-IDF cosine similarity follows the standard vector-space definition. Record linkage, survival estimation, proportional-hazards diagnostics, and change-point detection are supported by the original Fellegi–Sunter (1969), Davis–Goadrich (2006), Saito–Rehmsmeier (2015), Kaplan–Meier (1958), Cox (1972), Grambsch–Therneau (1994), and PELT (Killick et al., 2012) methods. Regime detection follows the Markov-switching framework of Hamilton (1989). The causal-inference outline in §18.1 draws on Angrist and Pischke (2009), Pearl, Glymour and Jewell (2016), Athey and Imbens (2017), and Callaway and Sant’Anna (2021). None of these sources validates this project’s own labels, numerical results, or thresholds; they justify method choices only. Full URLs and the specific design implications are recorded in `METHODOLOGICAL_REFERENCES.md`.

- [Official BOAMP API \(DILA\)](#).
- [INSEE SIREN definition](#) and [INSEE SIRET definition](#).
- [Commission Regulation \(EC\) No 213/2008 on CPV](#) and [Directive 2014/24/EU, Article 33](#).
- [Cosine similarity documentation](#).
- [Davis–Goadrich \(2006\)](#) and [Saito–Rehmsmeier \(2015\)](#) on ROC and precision-recall evaluation for skewed binary decisions.
- [Fellegi–Sunter \(1969\)](#), [Kaplan–Meier \(1958\)](#), and [Cox \(1972\)](#).
- [Grambsch–Therneau \(1994\)](#) and [Killick et al. \(2012\)](#).

- [Hamilton \(1989\)](#) on Markov-switching regime models, the framework underlying the trend HMM.
- [Angrist and Pischke \(2009\)](#), [Pearl, Glymour and Jewell \(2016\)](#), [Athey and Imbens \(2017\)](#), and [Callaway and Sant'Anna \(2021\)](#) on causal identification, cited only to support the outlined-not-executed causal-inference perspective.